

Report

Project Title:

Customer Churn Prediction & Retention Framework

Submitted By

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Introduction

Customer Churn Prediction & Retention Framework is a data-driven system designed to identify customers who are likely to stop using a company's products or services. Customer churn is one of the major challenges faced by businesses because losing customers can reduce revenue and increase the cost of acquiring new customers. By analyzing customer data such as usage patterns, subscription details, transaction history, and customer behavior, machine learning techniques can predict the probability of customer churn.

This project aims to develop a predictive framework that helps organizations detect potential customer loss at an early stage and implement effective retention strategies. The framework uses data analytics and machine learning algorithms to classify customers as likely to churn or remain loyal. The insights generated by the system enable businesses to improve customer satisfaction, enhance customer relationships, and increase profitability.

Problem Statement

Many organizations experience customer loss due to competition, dissatisfaction, poor service quality, or changing customer needs. Identifying customers who are likely to leave a service is difficult when dealing with large amounts of customer data. As a result, companies often fail to take timely actions to retain valuable customers.

The problem addressed in this project is to build a Customer Churn Prediction & Retention Framework that can analyze customer information, predict the likelihood of churn, and provide recommendations for customer retention. The proposed system helps businesses reduce customer attrition, improve customer loyalty, and make informed decisions using predictive analytics and machine learning techniques.

Data set Description

1. Loaded dataset using Pandas.
2. Checked for missing values.
3. Converted TotalCharges to numeric format.
4. Removed null and duplicate records.
5. Encoded categorical variables.
6. Verified data consistency.

4. Exploratory Data Analysis (EDA)

Several visualizations were generated to understand customer behavior.

Figure 1: Customer Churn Distribution (Pie Chart)

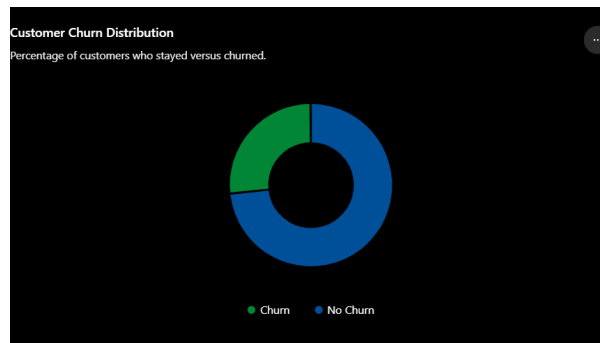


Figure 1: Enter Caption

Figure 2: Customer Churn Distribution

Description: This pie chart illustrates the proportion of customers who stayed with the company versus those who churned.

Observation: Approximately 73.5% of customers remained with the company, while 26.5% discontinued the service.

Inference: Although the majority of customers are retained, the churn percentage is significant and requires attention.

Figure 2: Contract Type Distribution (Pie Chart)

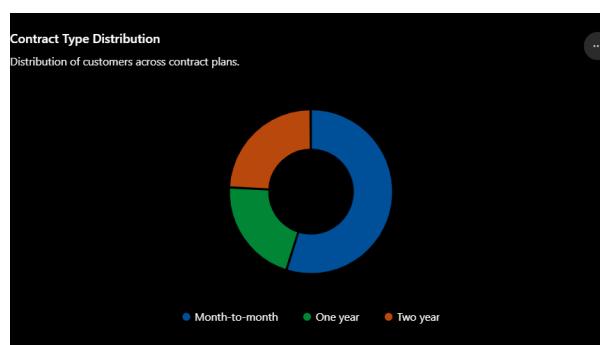


Figure 3: Enter Caption

Figure 4: Contract Type Distribution

Observation: Most customers prefer Month-to-Month contracts, followed by Two-Year and One-Year contracts.

Inference: Customers favor flexible contract options, but these plans are associated with higher churn rates.

Figure 3: Senior Citizen Distribution (Pie Chart)

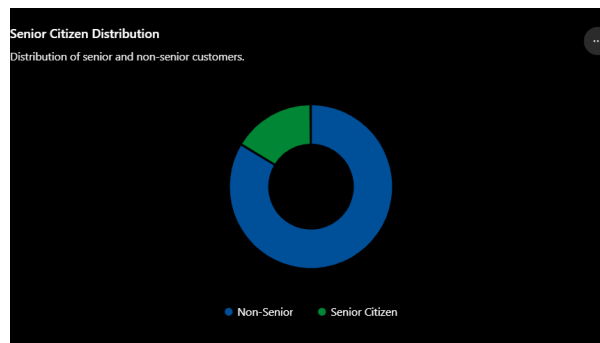


Figure 5: Enter Caption

Figure 6: Senior Citizen Distribution

Observation: Senior citizens represent a smaller portion of the customer base.

Inference: Although fewer in number, senior citizens tend to exhibit higher churn behavior.

Figure 4: Paperless Billing Usage (Pie Chart)

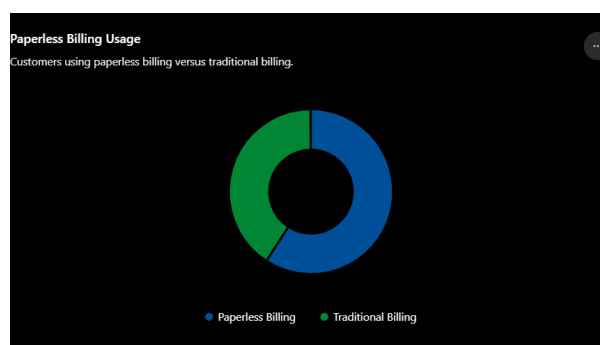


Figure 7: Enter Caption

Figure 8: Paperless Billing Usage

Observation: Most customers have adopted paperless billing.

Inference: Digital billing is widely accepted among customers and reflects modern service preferences.

Figure 5: Gender Distribution (Pie Chart)

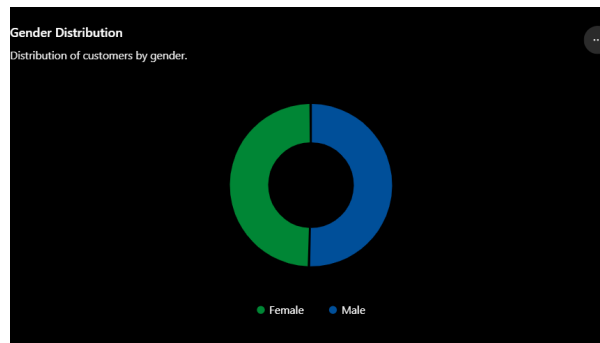


Figure 9: Enter Caption

Figure 10: Gender Distribution

Observation: The customer base is nearly equally divided between male and female customers.

Inference: Gender does not significantly influence customer enrollment.

Churn Analysis Graphs

Figure 6: Churn Rate by Contract Type

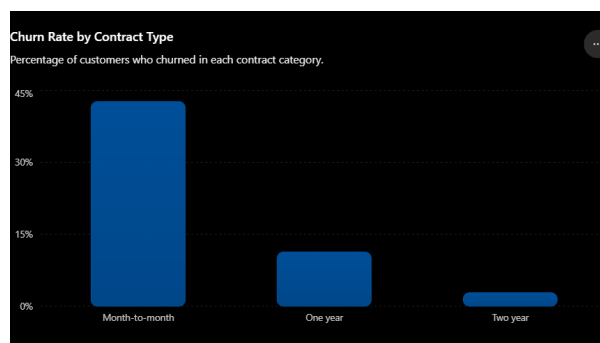


Figure 11: Enter Caption

Figure 12: Churn Rate by Contract Type

Observation: Month-to-Month customers have the highest churn rate, while Two-Year contract customers have the lowest.

Inference: Long-term contracts contribute significantly to customer retention.

Figure 7: Churn Rate by Internet Service

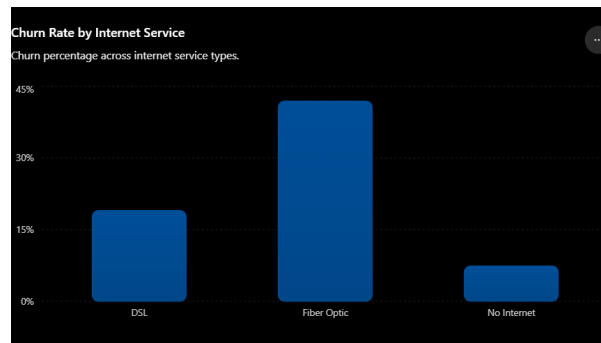


Figure 13: Enter Caption

Figure 14: Churn Rate by Internet Service

Observation: Fiber Optic users have the highest churn rate.

Inference: Service quality, pricing, or customer expectations may be influencing churn among Fiber Optic users.

Figure 8: Churn Rate by Payment Method

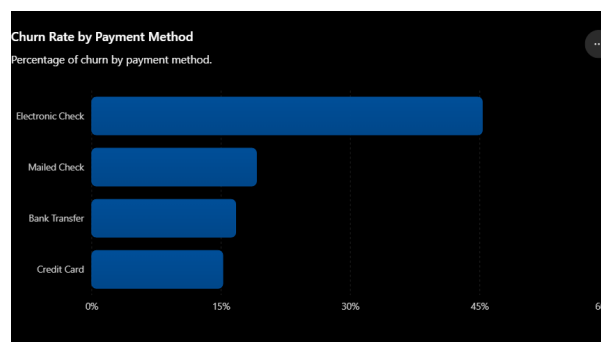


Figure 15: Enter Caption

Figure 16: Churn Rate by Payment Method

Observation: Electronic Check users exhibit the highest churn percentage.

Inference: Customers using electronic checks may be less committed or less satisfied with the service.

Figure 9: Churn Rate by Gender

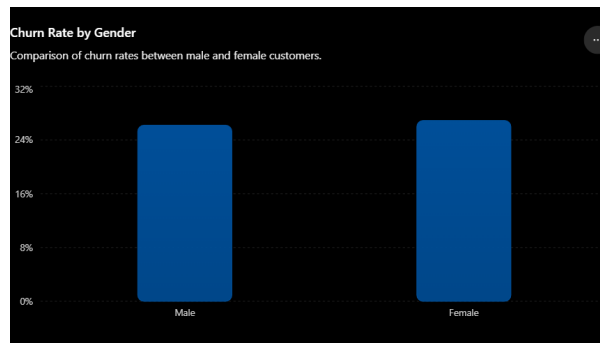


Figure 17: Enter Caption

Figure 18: Churn Rate by Gender

Observation: Both genders show nearly identical churn rates.

Inference: Gender is not a major factor affecting customer churn.

Figure 10: Churn Rate by Customer Tenure

Figure 19: Churn Rate by Customer Tenure

Observation: Customers within the first year have the highest churn rate.

Inference: Customer loyalty increases as the duration of service increases.